From Bridge Tables to the Negotiation Table: Cross-Domain Behavioral Consistency of Game-Derived Decision-Making Profiles

Anna Ben-Shushan and Jari Hämäläinen

Department of Computational Engineering, LUT School of Engineering Sciences, LUT University, Lappeenranta, Finland
anna.ben.shushan@student.lut.fi

Abstract. Business negotiation is one of the most consequential forms of decision-making under uncertainty, yet it is difficult to study empirically: real negotiations are private, and no public dataset links how a person makes decisions in general to how they negotiate. We propose contract bridge as a measurable laboratory. Bridge bidding is a structured sequence of decisions under incomplete information in which every action is recorded and scored, and the same player is observed across hundreds of comparable decision problems. From 149,208 board-level records of five European championships (2016–2025) we show, first, that elite players form a statistical *continuum* of styles rather than discrete clusters. We therefore identify the extremes of this continuum as four statistically validated decision-making profiles, and we turn each profile into an autonomous software agent whose behavior is grounded in the skills that a language model reads from that player's actual deals. The agents then play bridge and negotiate simulated business deals against fixed counterparts. Two findings emerge. Whether a profile wins in one domain says little about whether it wins in the other: winning depends on the opponent and on the rules of the game. How a profile behaves, however, carries over strongly: profiles that take risks at the bridge table make bolder demands at the negotiation table, and a control experiment that swaps the underlying skills between profiles reverses this relationship, showing that the behavior comes from the game-derived skills and not from the profile's name. The study is a validated feasibility result rather than a proof, and all data, code, and outputs are released so that every number in the paper can be reproduced.

Keywords: Decision-making under uncertainty · Contract bridge · Negotiation · Large language model agents · Behavioral profiling · Cross-domain transfer · Reproducibility.

1 Introduction

Negotiation, over prices, alliances, contracts, and platforms, is a sequence of decisions under uncertainty with conflicting objectives: demand more and risk the deal, or concede and lose surplus. Despite its economic weight, negotiation

behavior is hard to study at the level of the *individual decision-maker*. Real negotiations are private and rarely recorded; the transcripts that are public (teaching cases, mediation examples) are edited and anonymized; and, most importantly, no public dataset links the *same person's* general decision-making style to their observed negotiation behavior.

Contract bridge offers a way around this measurement problem. Bridge bidding is a formal negotiation-like protocol: partners exchange information through a constrained language of bids while opponents contest the auction, all under incomplete information (each player sees only 13 of 52 cards). Unlike real negotiations, *every* decision is recorded, timestamped to a specific deal, and objectively scored, and elite players are observed over hundreds of comparable decision problems. Bridge has long served as a benchmark for AI under imperfect information [1,2]; here we use it differently: not to play bridge well, but as a *measurable laboratory of individual decision-making style*.

This paper asks a single question:

> *If we derive a player's decision-making profile from their real bridge record and instantiate it as an autonomous agent, does the profile's behavioral signature persist when the agent negotiates in a business setting?*

We answer it with a four-stage pipeline (Secs. 4 to 7) and report both a positive result and an instructive negative one.

The paper makes five contributions, stated here in the order in which the reader meets them. The first is a negative result about the data itself: across five escalating preprocessing configurations and three algorithm families, elite championship players show no discrete cluster structure; they form a *continuum* of styles (Sec. 4.2). To our knowledge this is the first systematic demonstration of style-space continuity at championship level, and it redirects the methodology. The second contribution follows from it: we profile the *tails* of the continuum with a statistically gated rule, under which a player is assigned to a profile only if they lie in the top decile of the profile's defining axis and a one-sided binomial test rejects the population baseline (Sec. 4.3). We also document how expert review exposed a small-sample artefact in our first version and how the gate repairs it, a reusable recipe for profiling rare behaviors. The third contribution is an *evidence-grounded* way of turning a profile into an agent: a large language model (LLM) reads a seeded sample of each profiled player's raw deals and writes observable skills with per-deal evidence citations; it is directed to the profile's defining axis but never shown the engineered feature values, so every skill must be verifiable from the cited deals (Sec. 5). Fourth, we build a *validated cross-domain testbed* in which profile-conditioned agents play bridge and negotiate against fixed, rule-based counterparts; the bridge metric is validated with a zero-intelligence baseline [13] and a double-dummy reference [1], and the negotiation counterpart is calibrated on 5,247 human bargaining dialogues [7] (Sec. 7). The fifth contribution is the principal finding. Decision *style* transfers strongly (behavioral alignment $\rho_S = +0.80$), and an inverse-prompt control that swaps the extracted skills between profiles flips the correlation to $\rho_S = -0.90$,

demonstrating that the transfer is carried by the skills. Whether a style *wins* in both domains is weak and metric-dependent ($\rho_S \approx +0.2$ to $+0.5$): winning depends on the opponent and the payoff structure, behaving does not (Sec. 9).

Throughout, we adhere to two methodological principles: every experimental condition needed to reproduce the study is stated explicitly (Appendix A maps each reported number to a runnable script), and evaluation scenarios are ordered from *trivial*, where the reader can predict which profile should prevail and so builds trust in the instrument, to non-trivial.

The rest of the paper is organized as follows. Section 2 places the work in the literature on bridge AI, negotiating agents, and the psychology of behavioral consistency. Section 3 describes the corpus. The pipeline is then presented in four stages, each a section of its own: Stage 1 (Sec. 4) derives behavioral features from the records and discovers the decision-making profiles; Stage 2 (Sec. 5) has a language model read each profiled player's deals and write down the observable skills; Stage 3 (Sec. 6) builds an agent from each profile; and Stage 4 (Sec. 7) evaluates the agents in bridge and in negotiation. Section 8 describes how the pipeline is engineered as research software. Section 9 reports the results, Section 10 interprets them and states the limitations, and Section 11 concludes with future work. Everything needed to reproduce the results is collected in Appendix A.

2 Related Work

Bridge has long served as an *AI benchmark*. Ginsberg's GIB introduced double-dummy analysis as an objective reference for bridge decisions [1], and Rong et al. trained estimation and policy networks for competitive bidding on about one million deals [2]. These works optimize play strength; we instead treat the human record as behavioral data.

A second line of work builds *game-playing and negotiating agents*. Self-play systems master perfect-information games [3]; Diplomacy work combines language and strategic reasoning and studies honesty and sanctioning in negotiation-like play [4,5]. End-to-end negotiation dialogue agents were introduced by Lewis et al. [6]; He et al. decoupled strategy from generation and released the Craigslist Bargaining corpus we use for calibration [7]. Human negotiation corpora with per-participant trait measures now exist, notably CaSiNo, a multi-issue corpus with Big-Five and social-value-orientation scores per negotiator [11]. For LLM negotiators, NegotiationArena benchmarks multi-turn agent-vs-agent bargaining and shows that injected behavioral tactics measurably shift payoffs [9]; Abdelnabi et al. contribute multi-party, multi-issue scorable negotiation games [10]; and a 180,000-negotiation competition finds that prompted *warmth* and *dominance* systematically shape outcomes [8], independent evidence that prompt-carried style becomes bargaining behavior. Our contribution is orthogonal to all of these: we do not train or benchmark a negotiator; we test whether a profile derived from *real cross-domain human data* persists in another domain.

The *zero-intelligence* tradition supplies our null model. Gode and Sunder showed that budget-constrained random traders ("ZI-C") achieve high allocative efficiency, making them the canonical null model for market behavior [13]. We adopt a ZI-C-style bidder as the falsification baseline for our bridge skill metric: any metric that a constrained random agent can top is measuring structure, not skill.

Our profiling method is a relative of *archetypal analysis*, which represents data by its extreme points rather than by central prototypes [15]; our tail profiling operates on individual, interpretable behavioral axes with an explicit significance gate, rather than fitting a convex-combination factorization. From behavioral economics, prospect theory motivates why payoff structure shapes risk appetite [14], coopetition frames the simultaneous cooperation and competition that bridge partnerships embody [16], and classical negotiation analysis provides the surplus-splitting lens [17,18].

Whether behavioral dispositions persist across situations, the question of *cross-situational consistency*, is a classical one in personality psychology: consistency emerges reliably only when behavior is *aggregated* over many observations [31], and stable dispositional signatures coexist with systematic situational variation [32]. For risk attitude specifically, human self-report evidence indicates substantial domain-specificity [33]. These findings calibrate the scope of our claim: we measure the cross-domain consistency of *profile-conditioned agents*; whether the same consistency holds for the underlying human players is a distinct, open question left for future work with trait-annotated human negotiation data (Sec. 11).

Finally, language models are increasingly used as *behavioral simulacra*: conditioned on personas, they can reproduce human-like behavioral distributions [26,27,28]. Two validity risks are documented. The first is *tautology*: an agent told it is aggressive will act aggressively. The second is the converse, *non-enactment*: role-playing agents can state a persona's beliefs yet fail to behave accordingly [12], so persona conditioning cannot simply be assumed to work. Our design answers both empirically: skills are extracted from raw hands under a deal-level evidence requirement (Sec. 5), and an inverse-prompt control swaps skills between identities and measures whether behavior follows (Sec. 9.3), precisely the belief-vs-behavior consistency check this literature calls for.

3 Data: 149,208 Boards from Five European Championships

The corpus was scraped from the public tournament archive of the European Bridge League (EBL) for five championships: Budapest 2016, Ostend 2018, Madeira 2022, Herning 2024, and Poznan 2025. The categories covered are Open, Women, and Senior in 2016 and 2018; Open, Women, Senior, and Mixed in 2022 and 2024; and Open and Mixed in the 2025 archive. The unit of record is one board (deal) played in one room. In team play every board is played twice, in an Open and a Closed room, by different pairs holding the same cards, so

each deal yields two independent auctions and results; this duplicate structure is what later gives us an opponent-controlled skill measure (Sec. 7.1). The corpus holds **149,208 board-room records** with 48 fields each and about 2,500 named players (Table 1).

Table 1. Corpus composition and field coverage per championship.

Championship	Rows	Player names	Card holdings	Bidding sequences
Budapest 2016	38,240	100%	56%	0%
Ostend 2018	32,384	100%	53%	0%
Madeira 2022	32,256	100%	51%	0%
Herning 2024	34,048	100%	56%	100%
Poznan 2025	12,280	100%	51%	100%
Total	149,208	99.9%	54%	31% (46,230 sequences)

Each record carries four groups of fields: identity (championship, category, match, round, board, room); the four players seated in that room, recovered from the published seating diagrams; the deal itself, that is, the 52 cards by seat, where the EBL published them; and the auction, followed by the contract, the declarer, a doubled marker, the number of tricks taken, and the score. The auction is stored as a string of seat-labeled calls. Each item reads `seat:call`, where the seat is N, E, S, or W and the call is `Pass`, a bid (a level from 1 to 7 followed by the suit: `C` clubs, `D` diamonds, `H` hearts, `S` spades, or `NT` for no trump), or `x` for a double. Calls are listed four to a row starting from West, rows are separated by `|`, and `-` marks a seat whose turn came before the dealer's. One record, taken verbatim from the corpus, reads (dealer East, all vulnerable):

```
W:- N:- E:Pass S:1C | W:1D N:1S E:Pass S:2C |
W:Pass N:3NT E:Pass S:Pass | W:Pass
```

Read from left to right: East passed, South opened one club (`1C`), West overcalled one diamond (`1D`), North bid one spade, East passed, South rebid two clubs, West passed, North jumped to three no trump (`3NT`), and the other three players passed, which ends the auction. This is Board 10 of a Round-5 match in the Mixed category at Herning 2024, Open Room: North played 3NT, took nine tricks, and scored +600. Two quality flags mark whether the auction and the cards are present, and every analysis filters on the flag it needs rather than discarding partial records.

Coverage differs by championship for reasons of publication, not of play. Older EBL microsites do not expose machine-readable auctions, so bidding-dependent features use the 46,230 records (31%, Table 1) of Herning 2024 and Poznan 2025, while contract and outcome features use all 149,208; card holdings are published for 54% of boards. Cleaning is light and mechanical: contract strings are parsed into level, strain, and doubled status; a contract is scored as made when the

tricks taken reach the level plus six; and the declarer's name is resolved from seat and room. Three things the corpus does not contain are worth stating, because they bound what the paper can claim: the card-by-card play of each hand, the time taken over each call, and the partnership's bidding system. All data are public tournament records of named professional players published by the EBL; we collect no private or demographic information.

4 Stage 1: Discovering Decision-Making Profiles

4.1 Behavioral Features

For every player we compute per-player rates from two complementary families: *outcome* features, derived from the final contract the player declared (what they committed to), and *process* features, derived from their calls inside the auction (how they got there). A player enters the analysis only with $n_{\text{declared}} \geq 50$ declared boards *and* $n_{\text{bidding}} \geq 50$ boards with auction data, leaving **563 qualifying players**. Interpreting per-player rates only after aggregating ≥ 50 observations follows the aggregation principle of behavioral-stability research [31]: single-situation behavior is noisy, but aggregated tendencies are stable.

Table 2. Behavioral features. Family A is computed on declarer boards (all 149K rows); Family B on any seat from full auctions (46K rows). Features with coefficient of variation cv < 0.10 across players (e.g. mean contract level, calls per board) carry almost no between-player signal and are excluded from structure analysis.

Family	Feature	Definition (per-player rate)
A (outcome)	`slam_rate`	declared contracts at level 6–7
A	`partscore_rate`	declared contracts at levels 1–3 (stopping low; a level-based proxy that counts 3NT as a stop)
A	`nt_rate`	declared contracts in no-trump
A	`double_rate`	own declared contracts doubled by opponents
B (process)	`opening_rate`	boards where the player opened the auction
B	`preempt_rate`	openings at level ≥ 2 (obstructive; also counts the strong 2♣ and 2NT openings)
B	`intervention_rate`	entries into the opponents' auction
B	`penalty_double_rate`	boards where the player doubled the opponents[1]

Note the asymmetry: `double_rate` is partly opponent-dependent (being doubled requires the opponents' decision) and is therefore *not* used as a profile-defining axis below; all four defining axes are the player's own actions.

[1] The feature counts all doubles (takeout and penalty alike); both are an active choice to contest the auction, so it measures *aggressive doubling*. We keep the historical column name for reproducibility.

4.2 Finding 1: A Continuum, Not Clusters

To ensure that the absence of cluster structure is not an artefact of inadequate preprocessing, we ran a five-configuration audit before accepting any negative result: (V1) StandardScaler only; (V2) StandardScaler + PCA(3) [38]; (V3a–c) RobustScaler + cv-filter + correlation pruning + Mahalanobis outlier removal ($\alpha = 0.01$) + PCA(5), feeding K-Means [36], a Gaussian mixture with BIC selection [37], and HDBSCAN.

Table 3. Cluster-structure audit. Silhouette follows Rousseeuw [20]; HDBSCAN follows [21].

Config	Algorithm	Best k	Silhouette	HDBSCAN clusters
V1	K-Means	4	∼0.15	0
V2	K-Means (PCA 3)	4	**0.24**	0
V3a	K-Means (full prep.)	2	0.17	n/a
V3b	GMM + BIC	2	0.14	n/a
V3c	HDBSCAN	n/a	n/a	**0**

Three observations converge. First, silhouette never exceeds 0.24 (values near 0 mean points sit on cluster boundaries). Second, the density-based HDBSCAN, free of K-Means' convexity assumptions, labels *all* 563 players as noise in every configuration. Third, *more* aggressive preprocessing makes separation *worse* (V2→V3a: 0.24 → 0.17): outlier removal deletes precisely the extreme players who carry the weak signal. The PCA spectrum decays gradually (PC1 = 24.6%, PC2 = 19.1%, PC3 = 13.5%, …) with no dominant component, the signature of a smooth distribution. We conclude that elite championship players occupy a **continuum** of styles, consistent with expertise research on maximal adaptation to task constraints among top performers [25]. Partitioning such data is the wrong operation; profiling its *tails* is not.

4.3 Extreme-Percentile Profiling with a Significance Gate

Let $x_a(i)$ be player i's rate on defining axis a, with population mean μ_a, standard deviation s_a, and z-score $z_a(i) = (x_a(i) - \mu_a)/s_a$. The four defining axes and their profiles are: `slam_rate` → *Slam Hunter*, `partscore_rate` → *Insurance Player*, `penalty_double_rate` → *Fighter*, and `nt_rate` → *NT Specialist*. Player i is assigned to profile $a^* = \arg\max_a z_a(i)$ iff both gates hold:

$$z_{a^*}(i) \;\geq\; Q_{0.90}\big(z_{a^*}\big) \qquad \text{and} \qquad \Pr\big[X \geq k_i \,\big|\, X \sim \mathrm{Bin}(n_i, \mu_{a^*})\big] < 0.05, \quad (1)$$

where k_i is the player's raw event count and n_i the relevant denominator (declared boards for Family A axes; bidding boards for the Fighter axis). Players failing either gate are *Generalists*, retained as the elite baseline, not discarded.

Profile names are human-chosen domain labels for "extreme on axis a"; nothing downstream depends on the name (Sec. 9.3 tests this).

The binomial gate exists because of an *expert-review episode* that is worth recording. Our first release (v2.0) used $n_{\text{declared}} \geq 20$ and the percentile gate alone, yielding 807 players and 64 Slam Hunters with a mean slam rate 2.8× baseline. Expert review by a national-level bridge player flagged a rare-event artefact: with a ∼4–6% slam baseline, a player with 20 boards and 2 slams shows a 10% rate whose exact 95% confidence interval, [1.2%, 31.7%] (Clopper–Pearson [39]), is indistinguishable from the baseline. Indeed 41% of v2.0 "Slam Hunters" had fewer than 30 declared boards. Version 2.1 raised the floor to 50 boards and added the binomial gate of Eq. (1): the cohort shrinks to 20 Slam Hunters whose median n_{declared} jumps from 42 to **216** and whose CI, $[6.3\%, 15.2\%]$, clears the baseline. Effect ratios become more conservative (e.g. 2.8× → 1.84×), but every surviving assignment is statistically defensible. We report this revision openly as a reusable methodological recipe: *for low-base-rate behaviors, a percentile cutoff alone manufactures false profiles; gate it with a significance test against the population rate.*

Table 4. Final profile assignment (v2.1; 563 players). "Ratio" is the profile's mean on its defining axis over the Generalist mean.

Profile	n (%)	Defining axis	Profile mean	Generalist mean	Ratio	Median n_{declared}
Slam Hunter	20 (3.6%)	`slam_rate`	0.101	0.055	1.84×	216
Insurance Player	21 (3.7%)	`partscore_rate`	0.684	0.570	1.20×	97
Fighter	37 (6.6%)	`penalty_double_rate`	0.131	0.085	1.55×	153
NT Specialist	17 (3.0%)	`nt_rate`	0.385	0.282	1.36×	114
Generalist	468 (83.1%)	(baseline)	n/a	n/a	n/a	117

4.4 Behavioral Validation of the Profiles

Are the profiles real behavior or a pipeline artefact? For each profile we measured its defining rate directly from the raw records (no clustering, no LLM) for a validation sample of five players per profile against the Generalist baseline. Because Generalists are themselves elite players, this elite-vs-elite comparison is conservative. The ratios here differ from those of Table 4 because the sample differs: there, the full profile cohorts; here, the five source players per profile whose hands feed Stage 2. All four profiles separate with very large effects (Table 5); Fig. 1 shows the behavioral fingerprints. Given the five-player validation samples, we read these d values as descriptive of separation magnitude rather than as precise population estimates; the inferential weight is carried by the per-player binomial gates of Sec. 4.3.

Table 5. Profile validation on raw data (one-sided Mann–Whitney U [23]; Cohen's d [22]; $d \geq 0.8$ is conventionally "large").

Profile	Defining metric	Ratio vs. Generalist	Cohen's d	p
Fighter	`penalty_double_rate`	1.31×	2.13	0.016
Insurance Player	`partscore_rate`	1.24×	3.30	0.004
Slam Hunter	`slam_rate`	1.37×	2.81	0.004
NT Specialist	`nt_rate`	1.27×	4.62	0.004

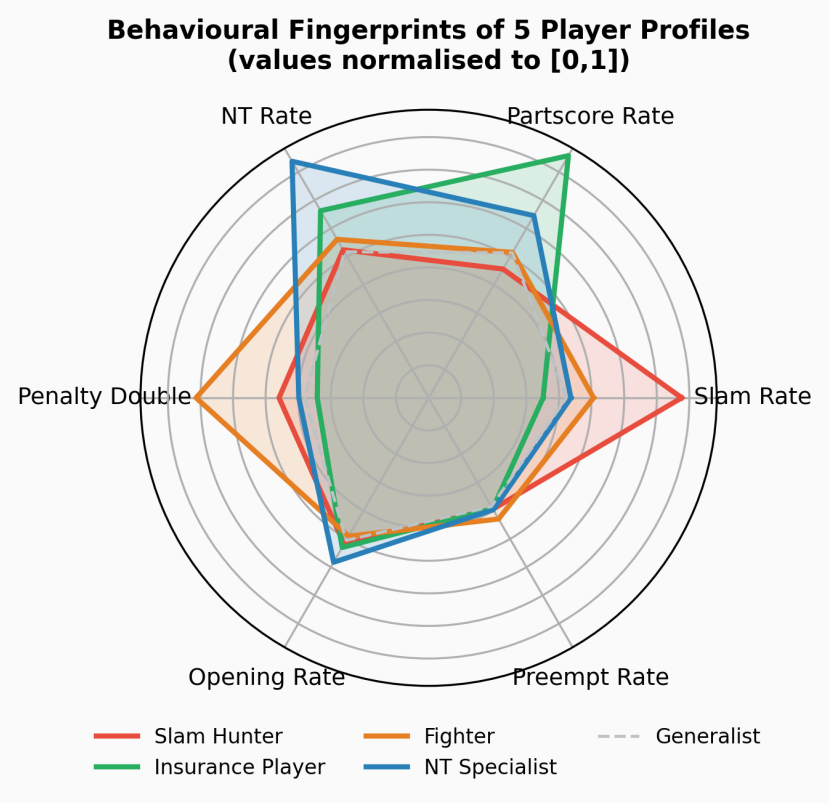


Fig. 1. Behavioral fingerprints of the four extreme profiles against the Generalist baseline (each feature min–max normalized to $[0, 1]$ across all players). Each profile is extreme on its own defining axis, and only mildly deviant elsewhere.

Several *known confounds* are retained as limitations. Bridge is a partnership game: part of an elevated slam rate may reflect an aggressive partner rather than the individual. The NT rate partly reflects the pair's bidding system (strong-club systems such as Precision open 1NT on a different point range). Two features are level-based proxies rather than strict bridge categories, and we keep them as com-

puted for reproducibility: `partscore_rate` counts 3NT, a game contract, among the low stops, and `preempt_rate` counts the strong 2♣ and 2NT openings, about a quarter of all openings at the two level or higher in this corpus, alongside the obstructive ones. The Insurance Player signal is the weakest (1.20×) and may partly reflect hand-distribution variance. The Fighter is methodologically the cleanest profile: doubling is a deliberate individual act, its denominator counts both sides of the table, and its base rate (∼8.5%) narrows confidence intervals quickly.

5 Stage 2: Evidence-Grounded Skill Extraction

For each profiled player we sample up to 50 boards (deterministic, seed 42) and format them in chunks of 25 boards per LLM call. Each board is rendered as four compact lines, showing the player's 13 cards, the full auction with seats labeled, the final contract, and the numeric result. A real board from the corpus (Herning 2024, Women, Round 19, Board 28) renders as:

```
Board 28 (Dealer West, Vul: N-S)
You are S (Declarer). Your hand: ♠AK752 ♡6 ♢AKQ ♣QT74
Bidding: W:Pass N:Pass E:Pass S:1S | W:Pass N:4C E:Pass S:4NT |
         W:Pass N:5C E:x S:Pass | W:Pass N:5H E:Pass S:6C |
         W:Pass N:6H E:Pass S:6S | W:Pass N:Pass E:Pass
Result: 6S by S, made 12 tricks, NS +1430
```

The extraction model (`gemini-2.5-flash`, temperature 0.3, JSON-schema output) is instructed to act as a bridge expert and return 3–5 *observable, falsifiable* skills, each with a name, a one-sentence behavioral description, per-board evidence citations, and a confidence grade; personality adjectives are explicitly forbidden. A profile-context block deliberately directs the model's attention to the profile's defining behavioral *axis* (e.g. "this player is in the top decile of slam bidders; at least two skills must concern slam bidding"); without such direction, rare defining behaviors drown under generic competitive-bidding observations. The model never receives the engineered feature values, the z-scores, or any comparative statistics: everything it asserts must cite specific deals as evidence, and Sec. 4.4 established, independently of any LLM, that the directed-to behaviors are genuinely present in these players' records. Whether downstream behavior follows skill content rather than profile identity is tested separately by the inverse-prompt control (Sec. 9.3). Skills are aggregated across chunks and players into a per-profile signature by TF–IDF cosine matching with a merge threshold of 0.40 (at 0.30, unrelated skills merged falsely; the threshold sweep is in the repository). TF–IDF is preferred over neural embeddings deliberately: at this scale (dozens of short, domain-specific skill names) lexical overlap is the relevant signal, and the aggregation stays deterministic and auditable, in line with the reproducibility goals. Typical extracted skills: *"uses Roman Key Card Blackwood to drive to slam with fitting hands"* (Slam Hunter); *"doubles opponents' partscores for penalty with trump stacks"* (Fighter); *"declines game invitations with borderline values"* (Insurance Player).

This design carries the first half of the paper's anti-tautology burden: the downstream agents are conditioned on *what the player demonstrably did with the cards*, not on adjectives we invented. The second half, that agent behavior follows this content rather than the profile's name, is carried by the inverse-prompt control (Sec. 9.3).

6 Stage 3: Profile-Conditioned Agents

Each of the five agents (four profiles plus the Generalist control) is an LLM (`gemini-2.5-flash`) whose system prompt is assembled from: (i) an identity line naming the profile and its one-line essence; (ii) the Stage-2 skill list, the only content derived from the player's actual record; (iii) domain rule constraints; and (iv) a mandatory JSON output schema with a free-text rationale field. The identity line does name the profile; whether measured behavior is driven by that name or by the skill content is exactly what the inverse-prompt control of Sec. 9.3 tests, by swapping only the skills while leaving every identity line in place. Bridge agents run at temperature 0.3 (later 0.0), negotiation agents at 0.7.

As a *mechanism check*, all five agents received the same two hands under the final prompts (Table 6; the archived outputs are in `experiments/`). The first is a flat 12-point hand (♠KJ95 ♡Q72 ♢KQ4 ♣J83) after the right-hand opponent opens 1♡: a competitive decision with no textbook answer. Here the call tracks the profile. The Insurance Player passes ("the safest option"), as does the Generalist ("not enough for a 1NT overcall, no five-card suit"), while the Fighter overcalls 1♠ on a four-card suit, "prioritizing competitive presence over strict suit-length requirements", and the Slam Hunter and the NT Specialist also act. The second is a 22-point hand with five-card support (♠AKQ72 ♡AK4 ♢A83 ♣Q2) after partner opens 1♠, where the rules of the game leave little room: four of the five agents make the textbook game-forcing raise (Jacoby 2NT), and only the Generalist stops in 3NT. Where the rules decide, the profiles leave little trace; where they do not, the profile decides. That is the intended behavior of an agent built as rules first, style second.

Table 6. Same hands, five agents (final prompts). Where the rules leave room (left), the call tracks the profile; where they do not (right), the agents converge.

Agent	12 HCP, after RHO's 1♡	22 HCP, after partner's 1♠
Slam Hunter	1♠	2NT (Jacoby)
Fighter	1♠	2NT (Jacoby)
NT Specialist	1♠	2NT (Jacoby)
Generalist	Pass	3NT
Insurance Player	Pass	2NT (Jacoby)

An expert pass over the *first* outputs on these hands had exposed bridge-mechanics errors: on the slam hand the Slam Hunter announced a "splinter" (a bid that promises a singleton) while holding a doubleton, and on the competitive hand an agent overcalled 1NT on 12 HCP. We hardened the prompts with three rule blocks (convention correctness, seat awareness, self-count of HCP) and reran. The convention-level errors disappeared; the price is the convergence visible on the slam hand, where the hardened agents all find the same correct call. Both prompt versions and all outputs are in the repository.

A note on the *reproducibility of LLM agents* is due here. Even at temperature 0 the API is not bit-reproducible (parallel floating-point reduction flips borderline decisions). We therefore (a) persist every raw model output, so all analyses are exactly reproducible from the archived JSON even if regeneration differs; and (b) report distributions over many decisions, never single calls. We flag this as a general methodological requirement for LLM-agent experiments: *sample-level, not call-level, reproducibility.*

7 Stage 4: Dual-Domain Evaluation

Both domains use *fixed, deterministic* counterparts, so that all between-agent variation is attributable to the profiles.

7.1 Bridge: Metric Falsification and a Real-Data Skill Measure

Our first bridge score rewarded bidding the level implied by combined high-card points. We falsified it with a **ZI-C–style baseline** in the spirit of Gode and Sunder's zero-intelligence constrained traders [13]: a bidder drawing uniformly from the legal calls *beat the elite-derived agents* on this metric: the score rewarded stating the formula, not making the contract. Any metric a constrained random agent can top is broken; we discarded it.

The corrected measure uses the duplicate structure of the data itself (no simulation). Every board is played at two tables; define the board *datum* as the cross-table mean score. A pair's skill on a board is its score minus the datum, converted to IMPs (the official non-linear scale that compresses extreme boards); player skill averages over a player's boards, profile skill over its players. Defensive excellence, including successful penalty doubles, scores automatically. On this scale the ZI-C bidder sits near -7 IMP/board, elite players near 0, and a double-dummy oracle [1] at $+1.1$ (Fig. 2); the metric now orders random $<$ human $<$ omniscient, as any valid skill measure must.

7.2 Negotiation: Protocol, Calibration, and a Trivial Example First

Each agent, acting as *buyer*, negotiates four business scenarios (acquiring a SaaS startup; buying out a co-founder's equity; acquiring a competitor's customer book; and a multi-year enterprise licence), three runs each, against one fixed rule-based seller. The seller opens at its ask P_{ask}, accepts any offer $\geq 0.95\,P_{\text{ask}}$,

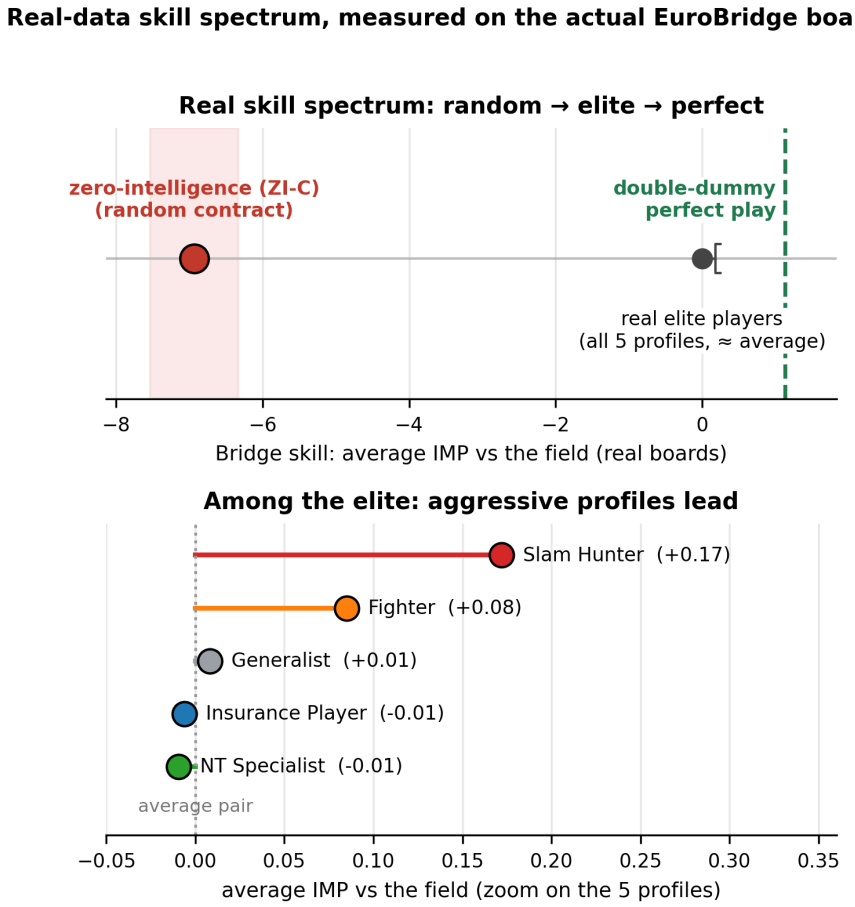


Fig. 2. The validated skill spectrum: the ZI-C random baseline loses ∼7 IMP/board to the field; elite profiles cluster near the datum; the double-dummy oracle gains +1.1. A metric that cannot reproduce this ordering is measuring the wrong thing.

otherwise concedes 15% of the current gap per round, never below its floor P_{floor}, for at most 4 rounds; it walks away on the second "absurd" offer (below $0.70\,P_{\text{floor}}$). These constants are not free parameters: they were calibrated on **5,247 human–human bargaining dialogues** of the CraigslistBargain corpus [7]: crowdsourced negotiations between people over real Craigslist listings (the corpus' training split; 6,682 dialogues in total, of which 4,334 carry usable price fields). In that corpus aggressive lowballing captures more surplus while the deal rate barely drops (≈ 0.83 vs 0.86): human sellers tolerate hard bargaining and abandon only absurd offers. An earlier, harsher walk-away rule reversed the results by punishing all aggression; we document that sensitivity in Sec. 9.1. The negotiation outcome is the *surplus captured*, $\sigma = (P_{\text{ask}} - P_{\text{deal}})/(P_{\text{ask}} - P_{\text{floor}}) \in$

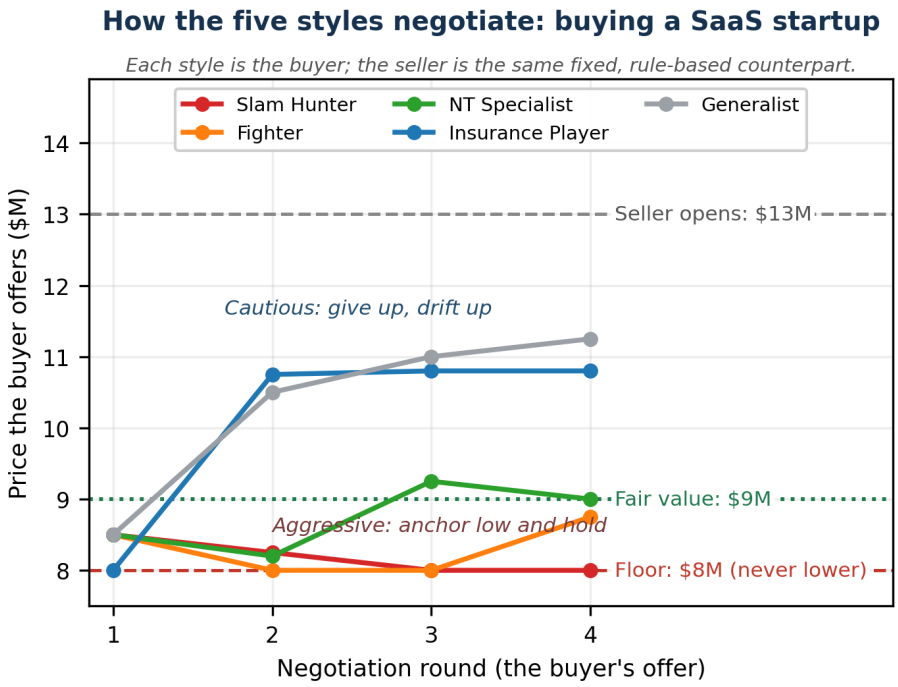


Fig. 3. One negotiation, five buyers (SaaS scenario: ask $13M, fair value $9M, floor $8M; real run-0 offer trajectories). Aggressive bridge profiles anchor low and hold; cautious profiles concede upward.

$[0, 1]$, zero if no deal. All four scenarios are single-issue price negotiations, *distributive* bargaining in the classical sense of Walton and McKersie [35]; integrative, multi-issue scenarios are deferred to future work (Sec. 11).

Following the principle of trivial-first evaluation, note the reader can predict this instrument's outcome: against a seller who concedes only under pressure, aggressive anchor-low profiles should capture more surplus, and cautious profiles should close early at worse prices. Fig. 3 shows one real scenario: the aggressive profiles anchor near the floor and hold; the cautious profiles drift up toward the seller. The instrument behaves as expected, which is what licenses its use on the non-trivial cross-domain question.

8 Implementation and Software Engineering

The pipeline is engineered as a single research software product rather than a collection of scripts, a choice worth stating explicitly given this volume's software-tools theme. All public operations flow through one SDK entry point (`src/sdk.py`); no function lives only in a notebook. LLM access is routed through

a provider-agnostic client (`src/shared/llm_client.py`) that logs model, token counts, latency, and cost for every call and enforces a budget guard (alert at $20, hard cap at $50); swapping the model provider is a one-argument change, which is how the cross-provider replication planned as future work is already wired. The code is typed, linted, and documented, with data hygiene enforced by convention: raw inputs are read-only, and all derived artifacts live under separate processed and results trees. A pytest suite of 178 tests covers the feature engineering, the profiling gates, the statistical validators, and the agents; LLM-dependent components are tested by injecting a mocked client, so the full suite runs without network access or cost. These practices are what make the reproduction protocol of Appendix A executable by a stranger rather than aspirational.

9 Results

9.1 Winning Does Not Transfer Cleanly

We first correlated per-profile *winning* in bridge with winning in negotiation (Spearman rank correlation over the $K = 5$ profiles). The answer depends on the bridge metric (Table 7): the discarded par-proxy gives $\rho_S = +0.30$; a fight-aware blend rises monotonically with the weight on doubling; the validated real-data IMP metric gives $\rho_S = +0.20$, with $+0.50$ under a raw-points robustness variant. With $K = 5$, none is significant.

Table 7. Winning↔winning is metric-sensitive. The final row is the validated metric of Sec. 7.1.

Bridge-side metric	ρ_S (win vs. win, $K = 5$)
Par-level proxy (falsified by ZI-C)	$+0.30$
Fight-aware blend, weight $w = 0.2/0.3/0.4$	$+0.60$ / $+0.70$ / $+0.90$
Real-data IMP-vs-datum (validated)	$+\mathbf{0.20}$

We read this as a substantive lesson, not a failure: *whether a style wins depends on the opponent and the payoff rules*: an over-harsh seller inverted the sign entirely in one ablation, so winning↔winning is the wrong invariant to chase across domains.

9.2 Style Transfers: Behavioral Alignment $\rho_S = +0.80$

The research question was always *behavioral* alignment. Define bridge-side aggression of a profile as the mean composite z-score of its members on the three action axes {`slam_rate`, `preempt_rate`, `penalty_double_rate`}, and negotiation-side aggression as *demand depth*: how far below the ask the agent's opening offer reaches,

$$\theta = \frac{P_{\text{ask}} - P_{\text{open}}^{\text{buyer}}}{P_{\text{ask}} - P_{\text{floor}}}, \tag{2}$$

averaged over scenarios and runs (from the archived logs; no new simulation). Demand depth operationalizes first-offer aggressiveness, whose anchoring effect on negotiation outcomes is well documented [34].

Table 8. Style↔style alignment across domains.

Profile	Bridge aggression (z)	Negotiation demand depth θ
Fighter	+1.30	0.808
Slam Hunter	+0.51	0.760
NT Specialist	−0.36	0.771
Generalist	−0.69	0.666
Insurance Player	−0.77	0.666
Spearman $\rho_S = +\mathbf{0.80}$ (two-sided $p = 0.10$, $K = 5$)		

Aggressive bridge profiles bargain aggressively; cautious ones bargain softly (Fig. 4). The one rank inversion (NT Specialist above Slam Hunter on θ) is what caps ρ_S at 0.80 rather than 1.0; this is expected, since the NT axis is stylistic rather than directly risk-graded.

A note on inference is due here rather than in the limitations. With $K = 5$, the permutation null admits only $5! = 120$ orderings, so the smallest attainable two-sided p is 0.017, and only near-perfect rankings ($|\rho_S| \geq 0.9$) clear the conventional 0.05 threshold; the observed +0.80 sits exactly one adjacent rank swap short of that, at $p = 0.10$. We therefore make no significance claim for the alignment itself and read +0.80 as an effect-size indication whose inferential support comes from converging evidence: the profile foundations are individually significant (Table 5, and every assignment gate of Eq. 1 holds at $p < 0.05$), and the inverse-prompt control (Sec. 9.3) is nominally significant on its own ($p = 0.037$).

9.3 The Inverse-Prompt Control: $\rho_S = -0.90$

Could the agents simply be acting out their names? Control: keep every identity line intact but inject the *opposite* profile's Stage-2 skills (Fighter gets the Insurance skill list and vice versa), then re-measure θ. If behavior follows the label, nothing should change; if it follows the skills, the correlation should invert. It inverts: the Fighter with Insurance skills softens ($\theta : 0.808 \to 0.725$), the Insurance Player with Fighter skills hardens ($0.666 \to 0.805$), and the cross-domain correlation flips to $\rho_S = \mathbf{-0.90}$ (two-sided $p = 0.037$; Fig. 5). The +0.80 alignment is therefore *skill-mediated*: it is carried by behavior extracted from real hands, not by the profile's name.

10 Discussion

The two findings are two sides of one coin. Decision *style*, how much risk you demand and how hard you hold a position, is a property of the decision-maker

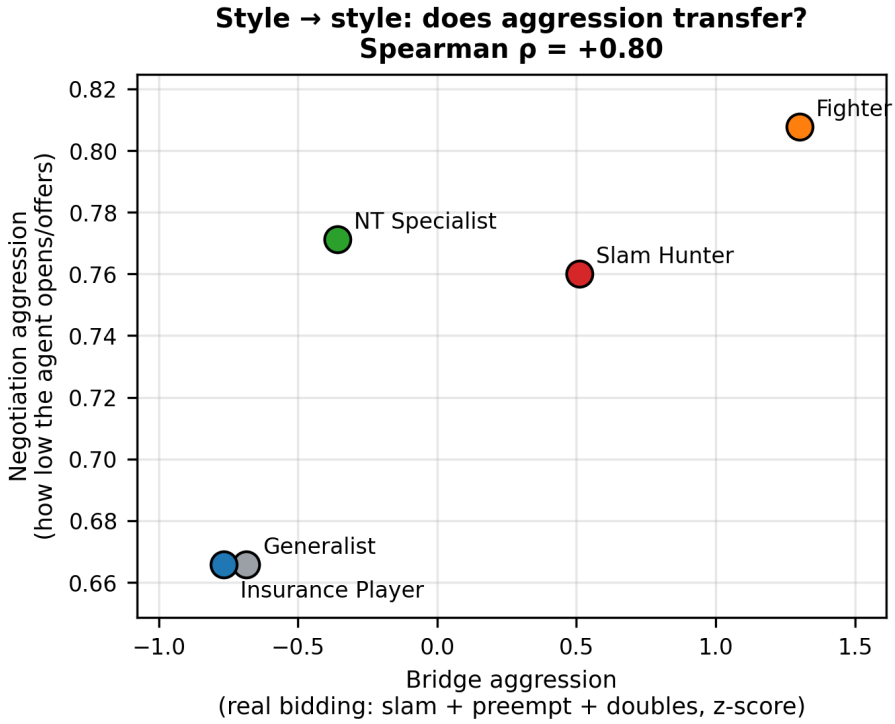


Fig. 4. Behavioral alignment: bridge-derived aggression (x) vs. negotiation demand depth (y), $\rho_S = +0.80$.

and survives domain transfer (+0.80, controlled). *Winning* is a property of the decision-maker *jointly with* the environment's payoff structure and counterpart, and does not transfer stably (+0.2 to +0.5, sign-reversible under an unrealistic counterpart). In optimization terms: the agents carry a stable preference structure over the risk–reward trade-off into a new decision problem, while the *value* of that preference structure depends on the objective landscape it meets. This distinction matters practically: profiling a counterpart's style from observable past behavior is feasible and predictive of *how* they will negotiate, not necessarily of who wins.

What do these findings mean beyond the laboratory? First, the continuum result argues that elite decision-making style is *dimensional* rather than typological: expert players differ by degree along a few interpretable axes, not by membership in discrete types. This parallels the move in personality psychology from types to traits, and it suggests that behavioral profiling in other expert domains should be built on graded axes with explicit significance gates rather than on clustering. Second, the transfer result says that what persists across domains is the *shape* of a decision-maker's risk preference, how much is demanded and

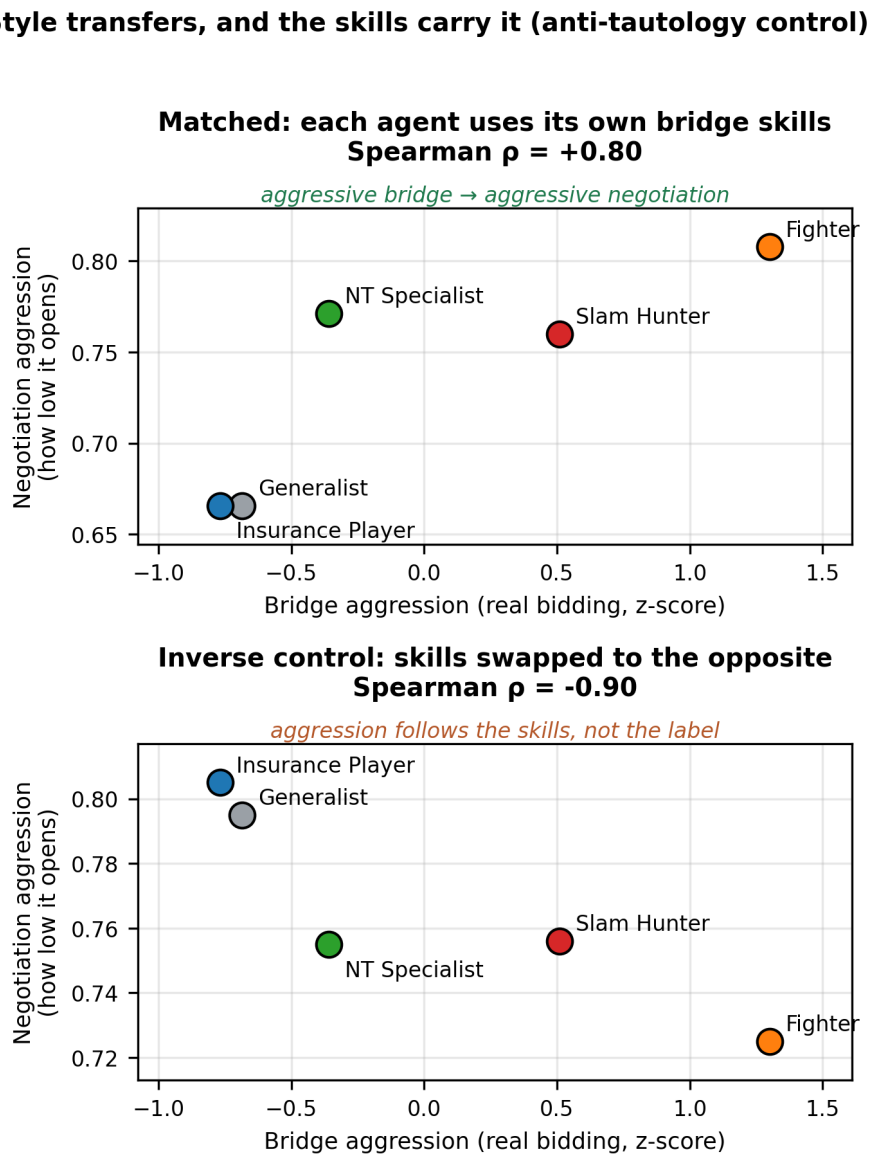


Fig. 5. Inverse-prompt control. Swapping each profile's bridge-derived skills (labels unchanged) flips the alignment from +0.80 to −0.90: the behavior follows the skills, not the label.

how firmly a position is held, rather than its payoff. For decision support this is the useful direction: a counterpart's past behavior in a structured, recorded domain is informative about how they will open and concede, even when it says little about who will win. Third, the control result shows that a language-model agent carries such a preference structure faithfully when, and only when, it is given behavioral content extracted from real records; the name of the profile contributed nothing measurable. That is both a methodological warning and an opportunity: persona-based simulations are only as good as the evidence behind the persona.

Three components are *methodological exports*, reusable beyond bridge: (i) the significance-gated tail-profiling recipe for low-base-rate behaviors (Eq. 1); (ii) metric falsification by constrained-random baseline (a ZI-C agent must rank last, an oracle first) before any cross-domain claim; (iii) evidence-grounded skill extraction plus an inverse-prompt control as an anti-tautology standard for LLM-persona experiments.

Several *limitations* bound these conclusions. With $K = 5$ profiles, only near-perfect rank orderings ($|\rho_S| \geq 0.9$) can clear $p < 0.05$ (Sec. 9.2); the observed alignment ($p = 0.10$) is therefore an indication, and scaling the number of profiles (finer tail slices, per-player agents) is the direct fix. The negotiation domain is simulated: the counterpart is rule-based (though data-calibrated), the scenarios are single-issue price negotiations, and surplus is the sole outcome. Bridge-side confounds remain (partnership effects on `slam_rate`; convention systems on `nt_rate`). The skill extractor and the agents share one model family (Gemini), so provider-specific behavior cannot yet be ruled out; the provider-agnostic client of Sec. 8 makes cross-provider replication (Claude, GPT) a one-argument change, and it is the first robustness item of future work. Finally, elite players are a selected population; transfer from amateur play is untested.

11 Conclusions and Future Work

This paper asked whether a decision-making profile derived from a person's real bridge record persists when that profile, instantiated as an autonomous agent, negotiates in a business setting. Within the limits of a five-profile study, the answer is that the *style* persists and the *success* does not. Along the way the bridge record itself yielded a finding: elite decision styles form a continuum whose tails are behaviorally real (Cohen's d 2.1–4.6). Those tails, once instantiated as agents, keep their risk signature in a new domain ($\rho_S = +0.80$; inverse control -0.90), while their winning record does not carry over cleanly ($\rho_S \approx +0.2$ to $+0.5$). The pipeline that produced these results, from 149,208 tournament records to agents evaluated in two domains, is released in full so that the findings can be checked and extended.

Three extensions are planned. *Team negotiations:* multi-agent teams mixing profiles (e.g. a Fighter–Insurance buying team) to study intra-team dynamics, bridging to coalition and alliance formation. *Multi-stage deals:* a memorandum-of-understanding phase followed by substantive bargaining, testing whether style

expression differs by phase. *Multi-criteria negotiation:* price plus quality plus optional features, which recasts the interaction as a multiobjective optimization problem with heterogeneous preference structures, and connects directly to interactive multiobjective decision support. On the bridge side, the natural deepening is a learned estimation–policy architecture (ENN/PNN [2]) over the full corpus, and per-player (rather than per-profile) agents to raise K from 5 to hundreds. On the negotiation side, future work targets real human negotiation data. A concrete first step is the CaSiNo corpus [11]: 1,030 multi-issue human negotiations in which each negotiator carries measured Big-Five and social-value-orientation scores: exactly the trait→behavior linkage this paper constructs synthetically, now testable on humans (does measured trait-level aggression predict human demand depth the way bridge-derived profiles predict it in agents?). Its integrative, three-issue structure also supplies the multi-criteria scenarios for the extension above, closing the loop the laboratory opened.

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References

1. Ginsberg, M. L.: GIB: Imperfect information in a computationally challenging game. Journal of Artificial Intelligence Research **14**, 303–358 (2001)
2. Rong, J., Qin, T., An, B.: Competitive bridge bidding with deep neural networks. arXiv preprint arXiv:1903.00900 (2019)
3. Silver, D., et al.: A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. Science **362**(6419), 1140–1144 (2018)
4. Kramár, J., et al.: Negotiation and honesty in artificial intelligence methods for the board game of Diplomacy. Nature Communications **13**, 7214 (2022)
5. Meta Fundamental AI Research Diplomacy Team (FAIR), Bakhtin, A., et al.: Human-level play in the game of Diplomacy by combining language models with strategic reasoning. Science **378**(6624), 1067–1074 (2022)
6. Lewis, M., Yarats, D., Dauphin, Y., Parikh, D., Batra, D.: Deal or no deal? End-to-end learning of negotiation dialogues. In: Proc. EMNLP 2017, pp. 2443–2453 (2017)

7. He, H., Chen, D., Balakrishnan, A., Liang, P.: Decoupling strategy and generation in negotiation dialogues. In: Proc. EMNLP 2018, pp. 2333–2343 (2018)
8. Vaccaro, M., Caosun, M., Ju, H., Aral, S., Curhan, J. R.: Advancing AI negotiations: A large-scale autonomous negotiation competition. Proceedings of the National Academy of Sciences **123**(23), e2521774123 (2026)
9. Bianchi, F., Chia, P. J., Yuksekgonul, M., Tagliabue, J., Jurafsky, D., Zou, J.: How well can LLMs negotiate? NegotiationArena platform and analysis. In: Proc. ICML 2024 (2024)
10. Abdelnabi, S., Gomaa, A., Sivaprasad, S., Schönherr, L., Fritz, M.: Cooperation, competition, and maliciousness: LLM-stakeholders interactive negotiation. In: Proc. NeurIPS 2024, Datasets and Benchmarks Track (2024)
11. Chawla, K., Ramirez, J., Clever, R., Lucas, G., May, J., Gratch, J.: CaSiNo: A corpus of campsite negotiation dialogues for automatic negotiation systems. In: Proc. NAACL 2021, pp. 3167–3185 (2021)
12. Mannekote, A., Davies, A., Li, G., Boyer, K. E., Zhai, C., Dorr, B. J., Pinto, F.: Do role-playing agents practice what they preach? Belief-behavior consistency in LLM-based simulations of human trust. arXiv preprint arXiv:2507.02197 (2025)
13. Gode, D. K., Sunder, S.: Allocative efficiency of markets with zero-intelligence traders: Market as a partial substitute for individual rationality. Journal of Political Economy **101**(1), 119–137 (1993)
14. Kahneman, D., Tversky, A.: Prospect theory: An analysis of decision under risk. Econometrica **47**(2), 263–291 (1979)
15. Cutler, A., Breiman, L.: Archetypal analysis. Technometrics **36**(4), 338–347 (1994)
16. Brandenburger, A. M., Nalebuff, B. J.: Co-opetition. Currency Doubleday, New York (1996)
17. Raiffa, H.: The Art and Science of Negotiation. Harvard University Press, Cambridge, MA (1982)
18. Nash, J. F.: The bargaining problem. Econometrica **18**(2), 155–162 (1950)
19. Shoham, Y., Leyton-Brown, K.: Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations. Cambridge University Press (2008)
20. Rousseeuw, P. J.: Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics **20**, 53–65 (1987)
21. Campello, R. J. G. B., Moulavi, D., Sander, J.: Density-based clustering based on hierarchical density estimates. In: Proc. PAKDD 2013, LNCS 7819, pp. 160–172. Springer (2013)
22. Cohen, J.: Statistical Power Analysis for the Behavioral Sciences, 2nd edn. Lawrence Erlbaum Associates, Hillsdale, NJ (1988)
23. Mann, H. B., Whitney, D. R.: On a test of whether one of two random variables is stochastically larger than the other. Annals of Mathematical Statistics **18**(1), 50–60 (1947)
24. Spearman, C.: The proof and measurement of association between two things. American Journal of Psychology **15**(1), 72–101 (1904)
25. Ericsson, K. A., Lehmann, A. C.: Expert and exceptional performance: Evidence of maximal adaptation to task constraints. Annual Review of Psychology **47**, 273–305 (1996)
26. Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., Bernstein, M. S.: Generative agents: Interactive simulacra of human behavior. In: Proc. UIST 2023 (2023)
27. Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., Wingate, D.: Out of one, many: Using language models to simulate human samples. Political Analysis **31**(3), 337–351 (2023)

28. Aher, G., Arriaga, R. I., Kalai, A. T.: Using large language models to simulate multiple humans and replicate human subject studies. In: Proc. ICML 2023 (2023)
29. Virtanen, P., et al.: SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods **17**, 261–272 (2020)
30. Pedregosa, F., et al.: Scikit-learn: Machine learning in Python. Journal of Machine Learning Research **12**, 2825–2830 (2011)
31. Epstein, S.: The stability of behavior: I. On predicting most of the people much of the time. Journal of Personality and Social Psychology **37**(7), 1097–1126 (1979)
32. Mischel, W., Shoda, Y.: A cognitive-affective system theory of personality: Reconceptualizing situations, dispositions, dynamics, and invariance in personality structure. Psychological Review **102**(2), 246–268 (1995)
33. Weber, E. U., Blais, A.-R., Betz, N. E.: A domain-specific risk-attitude scale: Measuring risk perceptions and risk behaviors. Journal of Behavioral Decision Making **15**(4), 263–290 (2002)
34. Galinsky, A. D., Mussweiler, T.: First offers as anchors: The role of perspective-taking and negotiator focus. Journal of Personality and Social Psychology **81**(4), 657–669 (2001)
35. Walton, R. E., McKersie, R. B.: A Behavioral Theory of Labor Negotiations: An Analysis of a Social Interaction System. McGraw-Hill, New York (1965)
36. Lloyd, S. P.: Least squares quantization in PCM. IEEE Transactions on Information Theory **28**(2), 129–137 (1982)
37. Schwarz, G.: Estimating the dimension of a model. Annals of Statistics **6**(2), 461–464 (1978)
38. Jolliffe, I. T.: Principal Component Analysis, 2nd edn. Springer, New York (2002)
39. Clopper, C. J., Pearson, E. S.: The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika **26**(4), 404–413 (1934)

A Reproducibility of Results

This appendix collects everything a reader needs to reproduce the numbers in the paper: the fixed constants and the software environment (Sec. A.1), and a numbered, end-to-end protocol that maps each table and figure to the script that produces it and states the expected output of every step (Sec. A.2).

A.1 Fixed Constants and Environment

All code, prompts, thresholds, seeds, and archived model outputs are in the public repository:

https://github.com/annafr2/From-Bridge-Tables-to-Global-Markets

The fixed constants are: qualification $n \geq 50$ declared and ≥ 50 bidding boards; percentile gate top 10%; binomial gate $\alpha = 0.05$ (one-sided, SciPy `binomtest` [29]); cv-filter 0.10; K-Means/GMM/HDBSCAN via scikit-learn [30] with `random_state`=42; LLM chunking 25 boards/call, ≤ 50 boards/player, sampling seed 42; skill-merge TF–IDF threshold 0.40; temperatures 0.3 (extraction), 0.3/0.0 (bridge), 0.7 (negotiation); seller rule: accept $\geq 0.95\,P_{\text{ask}}$, concede

15% of gap/round, floor-bounded, $\leq$ 4 rounds, walk-away after 2 offers below $0.70\,P_{\text{floor}}$.

The environment is Python 3.11 with scikit-learn, SciPy 1.11, and pandas (versions pinned in `requirements.txt`); all LLM stages use `gemini-2.5-flash` through a single provider-agnostic client. Every LLM call is logged with model, token counts, and cost; the total compute cost of all experiments was under $10.

A.2 Step-by-Step Protocol

The protocol below reproduces every table and figure in this paper. Steps 1–5 are fully deterministic (no API key, no cost). Steps 6–7 and 10 call the LLM and are therefore *distributionally* reproducible (Sec. 6); the raw model outputs of our runs ship in the repository, so steps 8–9 and 11 can be reproduced *exactly* from the archive without re-querying any model. Every command is run from the repository root.

1. **Environment.** Clone the repository (URL in Sec. A.1); create a Python 3.11 virtual environment; run `pip install -r requirements.txt`. For steps 6–7 and 10 only, set `GOOGLE_API_KEY` in `.env`.
2. **Data check.** Load the corpus via `src/shared/data_loader.py`.
   Expected: 149,208 board-room records; per-championship coverage as in Table 1.
3. **Stage 1: features and profiles.**
   Run the Stage-1 pipeline in `src/stage1_clustering/`: first `features.py` (with `MIN_BOARDS` = 50), then `extreme_profiles.py` (top-10% cutoff, $\alpha = 0.05$).
   Expected: `data/processed/player_profiles.csv` with 563 rows and profile counts 20/21/37/17/468 (Table 4).
4. **Cluster-structure audit.**
   `python -m notebooks.preprocessing_comparison`
   Expected: the five configurations of Table 3 (max silhouette 0.24; HDBSCAN 0 clusters) and `results/preprocessing.xlsx`.
5. **Profile validation.**
   `python -m notebooks.validate_base_rates`
   Expected: the effect sizes and p-values of Table 5.
6. **Stage 2: skill extraction** (LLM; ≈$1).
   Run the extractor in `src/stage2_skills/` over all profiled players: chunks of 25 boards, at most 50 boards per player, sampling seed 42, temperature 0.3, and the model `gemini-2.5-flash`; then aggregate with the TF–IDF merge threshold 0.40.
   Output: the per-profile skill signatures JSON.
7. **Stages 3–4: dual-domain simulation** (LLM; <$5).
   `python -m src.stage4_simulate.bridge_runner`
   `python -m src.stage4_simulate.negotiation`
   All raw agent outputs are persisted under `results/stage4/`.

8. **Bridge skill metric and spectrum.**
   `python -m notebooks.alignment_real_bridge`
   `python -m notebooks.visualize_real_skill_spectrum`
   Expected: Fig. 2 and the win–win row of Table 7 ($\rho_S = +0.20$ on IMPs, $+0.50$ on raw points).
9. **Style alignment.**
   `python -m notebooks.style_alignment`
   Recomputed from the archived Stage-4 logs with no new simulation. Expected, *exactly*: Table 8 and Fig. 4 ($\rho_S = +0.80$).
10. **Inverse-prompt control** (LLM; <$1).
    `python -m notebooks.inverse_prompt_control`
    Re-runs the swapped-skill negotiations with live model calls, so it reproduces distributionally; the archived outputs behind Fig. 5 ($\rho_S = -0.90$) ship in the repository.
11. **Seller calibration.**
    `python -m notebooks.validate_negotiation_features`
    Runs against the CraigslistBargain training split under `data/external/`. Expected: 4,334 usable dialogues; aggressive opening captures more surplus at a deal rate of ≈ 0.83 vs 0.86 (Sec. 7.2).